



Proximal Biceps Tendon Pathology

Samuel R.H. Steiner, John T. Awowale, Stephen F. Brockmeier

INTRODUCTION

Pathology to the proximal aspect of the long head of the biceps tendon (LHBT) has long been recognized as a source of shoulder pain and dysfunction.¹⁻³ Referred by some as the biceps-labral complex,⁴ the proximal biceps tendon includes its origin on the superior labrum, the intra-articular tendon, and the tendon as it courses its way through its stabilizing pulley down the bicipital tunnel to the biceps brachii muscle belly. For lesser abnormalities of the biceps tendon, such as tendinopathy, conservative treatment modalities are often successful. In the setting of superior labral pathology, partial LHBT tears, or instability of the biceps tendon in the bicipital groove, surgical intervention is an accepted treatment option after a conservative algorithm has failed. This chapter focuses on identifying proximal LHBT pathology based on a thorough history and physical examination in conjunction with imaging, and from there selecting the proper treatment. This chapter also reviews the current surgical techniques.

HISTORY

The structure of the glenohumeral joint makes diagnoses of LHBT pathology challenging. Other disease processes localized to the shoulder, such as rotator cuff disease, superior labral from anterior to posterior (SLAP) tears, glenohumeral joint arthritis, subacromial impingement, or acromioclavicular (AC) joint arthrosis, can complicate the differential diagnosis. Thus a thorough history and physical examination is needed to correctly diagnose LHBT disease.

Patients with suspected pathology of the long head of the biceps will present with progressive anterior shoulder pain and declining function as a result of chronic overuse. In many cases, this occurs from repetitive overhead activities. Patients who participate in baseball, softball, or volleyball may report ongoing clicking or snapping sensations with throwing or spiking motions. Alternatively, an isolated event may also be identified such as catching a heavy object with a flexed arm, resulting in an eccentric force causing an acute rupture. This is often described as an audible “pop” resulting in swelling of the arm, and ecchymosis with or without a visible defect commonly referred to as the “Popeye sign.” Pain is most commonly reported down the antero-medial aspect of the shoulder around the bicipital groove, sometimes traveling distally into the biceps muscle belly. In older patients with long head of the biceps pathology there is often

an associate rotator cuff tear present that is felt as pain along the anterolateral shoulder region.⁵

PHYSICAL EXAMINATION

The long head of the biceps originates at the supraglenoid tubercle and the superior glenoid labrum. It inserts distally along with the short head of the biceps on the bicipital tuberosity of the radius. The tendon is extrasynovial spanning the glenohumeral joint within the rotator interval (region anterior to the supraspinatus tendon and superior to the subscapularis tendon). The tendon sits in the bicipital groove, which is defined as the region between the greater and lesser humeral tuberosities. The bony anatomy and soft tissue structures help to maintain the long head of the biceps within the bicipital groove. As the tendon exits the bicipital groove it travels under the pectoralis major insertion, eventually combining with the short head of the biceps. The main blood supply to the long head of the biceps is provided by the branches of the anterior circumflex humeral artery.⁶

There are over 180 physical examination tests described for diagnosing pathology of the shoulder.⁷ As discussed earlier, proximal biceps tendon pathology is difficult to isolate on physical exam as many examination findings are not specific to proximal biceps tendon dysfunction alone (Table 46.1). The presence of the “Popeye sign” indicates a rupture of the long head of the biceps thus solving the diagnostic dilemma; however, this sign is often not present, necessitating the correct execution of physical examination maneuvers and interpretation of results to diagnose other proximal tendon pathology. Furthermore, patients with concomitant rotator cuff tears can make diagnosis of biceps tendon pathology via physical examination even more difficult.⁸ One of the most common findings is tenderness over the inter-tubercular groove of the proximal humerus showing up to 90% sensitivity in some studies for detecting long head of the biceps pathology.⁹ During this exam the practitioner palpates the bicipital groove along the anterior proximal humerus approximately 5 to 7 cm below the acromion with the arm internally rotated 10 degrees. The diagnosis of tendon pathology is more likely if pain during palpation moves laterally with continued palpation and external rotation (ER) of the arm. The asymptomatic contralateral arm should also be tested for comparison, as a normal biceps tendon also may be tender to deep palpation. Pathology of the biceps tendon under the proximal aspect of the pectoralis major muscle can be tested using the subpectoral bicep tendon

Abstract

Pathology to the proximal aspect of the long head of the biceps tendon (LHBT) has long been recognized as a source of shoulder pain and dysfunction. Referred by some as the biceps-labral complex, the proximal biceps tendon includes its origin on the superior labrum, the intra-articular tendon, and the tendon as it courses its way through its stabilizing pulley down the bicipital tunnel to the biceps brachii muscle belly. For lesser abnormalities of the biceps tendon, such as tendinopathy, conservative treatment modalities are often successful. In the setting of superior labral pathology, partial LHBT tears, or instability of the biceps tendon in the bicipital groove, surgical intervention is an accepted treatment option after a conservative algorithm has failed. This chapter focuses on identifying proximal LHBT pathology based on a thorough history and physical examination in conjunction with imaging, and from there selecting the proper treatment. This chapter also reviews the current surgical techniques.

Keywords

SLAP
biceps tendon
tenotomy
tenodesis
arthroscopic
biceps
tenodesis
open biceps

TABLE 46.1 Typical Physical Examination Tests for the Long Head Biceps Tendon and Concomitant Pathology

Pathology	Test	Specific Site	Examination	Positive
LHBT	Biceps instability ⁶⁵	LHBT within the groove	Full abduction, external rotation; palpate bicipital groove	Palpable “click”
	Point tenderness ⁶⁶	LHBT within the groove	Palpate the bicipital groove 3–6 cm below the acromion; internally rotate the arm 10 degrees	Reproducible pain; dynamic evaluation shows groove tenderness laterally as the arm is externally rotated
	Speed ⁶⁷	LHBT within the groove; SLAP	Elbow extended, forearm supinated, arm elevated to 90 degrees	Pain localized within the bicipital groove
	Yergason ⁶⁸	LHBT	With the elbow flexed and the forearm pronated, the examiner holds the arm at the wrist and the patient actively supinates against resistance	Bicipital groove pain
	Gerber’s lift-off test ⁶⁵	Subscapularis LHBT	The patient stands with his/her hand behind the back and the dorsum of the hand resting on the midlumbar spine; the patient attempts to raise his/her hand off the back by maintaining or increasing internal rotation of the humerus and extension of the shoulder	Inability to move the dorsum of the hand off the back
Concomitant RC	Belly press ⁶⁵	Subscapularis	The hand is on the abdomen and attempts are made to move it anteriorly	Difficulty moving the elbow forward
	Neer impingement ⁶⁹	AC joint	The arm is maximally passively elevated forward, with internal rotation with the scapula stabilized	Pain/weakness in the subacromial space/edge of the acromion of the biceps region
	Kennedy-Hawkins ⁵⁸	Impingement of the greater tuberosity and CH ligament	The arm is forward-flexed 90 degrees, then quickly rotated internally	Pain/weakness in deltoid or anterior shoulder
	Empty can ⁷⁰	Supraspinatus	The arm is forward-flexed 90 degrees, with full internal rotation 90 degrees; downward force is resisted	Pain/weakness deep in the shoulder
Concomitant labrum	Compression rotation ⁷¹	Labrum	The patient lies supine; the affected arm is elevated 90 degrees and the arm is rotated while an axial load is applied	Pain or clicking deep in the shoulder
	O’Brien’s active compression ⁷²	AC joint superior labrum	The patient stands with the arm adducted 15 degrees and forward-flexed 90 degrees with the elbow fully extended; the arm is maximally internally rotated and elevated with the palm up and the thumb pointing down against resistance	Pain in the AC joint and pain or a deep “click” in the GH joint
	Anterior slide ⁷³	Labrum	The patient stands with a hand on the hip of the affected side while the examiner applies an axial load along the humerus	Pain or a “click” is produced

AC, Acromioclavicular; CH, coracohumeral; GH, glenohumeral; LHBT, long head of the biceps tendon; RC, rotator cuff; SLAP, superior labral anterior to posterior.

test. The proximal biceps tendon is palpated underneath the pectoralis muscle just medial to its insertion. The arm is then internally rotated against resistance with pain indicating a positive test. Again, the contralateral normal shoulder also should be evaluated. A more painful response on the affected side validates the test result.¹⁰

Several other traditional physical examination techniques have been described to evaluate long head of the biceps. The Speed test is performed with forward flexion of the shoulder to 90 degrees with the forearm supinated and the elbow fully extended; pain with resisted shoulder flexion in this position is

a positive result. This test may also be positive in patients with SLAP lesions. The Yergason test is positive with pain to palpation over the bicipital groove during resisted supination of the forearm as the elbow is flexed at 90 degrees.⁵ Both the Speed and the Yergason tests have been shown to be less sensitive 27% to 40%, though more specific 80% to 100%.⁹ A less specific test, although more sensitive, is the O’Brien active compression test, which may indicate either long head of the biceps tendonitis, SLAP lesions, or AC joint arthrosis.⁹ This test is performed with the arm adducted 10 to 15 degrees across the body with the elbow extended. Pain with resisted forward flexion of the arm

in this position with the forearm pronated indicates a SLAP tear; however, pain with resisted forward flexion in this position with both pronation and supination indicates potential AC joint pathology.⁵ Newer tests for evaluating long head of the biceps pathology have also been developed, such as the “upper cut” maneuver.¹¹ This is performed with the involved shoulder at in neutral position with the elbow flexed to 90 degrees, forearm supinated, and the patient’s hand making a fist. The patient then attempts to forward flex the shoulder and elbow towards the chin in a punching-like maneuver as the practitioner holds the patient’s fist, resisting the motion. A positive test is seen with pain along the anterior shoulder during the resisted movement. The 3-Pack examination is also a newer test for evaluating for proximal biceps pathology (Fig. 46.1). The maneuvers that comprise the 3-Pack exam are the active compression test (O’Brien sign), the resisted throwing test, and the bicipital tunnel palpation. The 3-Pack examination demonstrated a higher sensitivity but lower specificity when compared to more traditional exam maneuvers such as the Speed test, Yergason test, full can test, and the empty can test.¹²

The biceps tendon is held within the bicipital groove by a tendoligamentous sling or pulley made up of the supraspinatus tendon, coracohumeral ligament, superior glenohumeral ligament, and notably the subscapularis tendon. Medial biceps tendon subluxations or dislocations occur when this pulley mechanism has been disrupted, most commonly with partial or complete tearing of the subscapularis tendon. On physical examination, medial biceps instability can be detected by a palpable click or tenderness to palpation of the biceps tendon on full abduction and ER of the affected arm. When the tendon is fully dislocated

it can be palpated anterior to the lesser tuberosity and rolled under the examiner’s finger, eliciting tenderness.⁵

Selective injections about the shoulder are a good way to distinguish various pathology as they can be used therapeutically and diagnostically.¹³ A subacromial space injection will alleviate pain with impingement. Pain relief with an injection into the area of the bicipital groove may be indicative of long head of the biceps tendonitis.⁹ Intra-articular glenohumeral joint injections are useful when evaluating a potential SLAP lesion.⁵

IMAGING

The diagnosis of biceps tendon pathology is often aided via imaging modalities. Plain radiographs with orthogonal views (AP, lateral, and axillary) should be obtained initially. Although pathology of the biceps tendon will usually not be apparent on plain radiographs, they are useful in ruling out other shoulder pathology such as glenohumeral joint degeneration or rotator cuff calcific tendonitis. To further evaluate the bicipital groove radiographically, a Fisk view can be obtained. This may show bicipital groove osteophytes or narrowing. The view is obtained with the patient supine with their arm at their side with the hand fully supinated, the cassette is placed on the top of their shoulder and the x-ray is directed cephalad down the bicipital groove.¹⁴ In certain patients for whom magnetic resonance imaging (MRI) is contraindicated, a plain arthrogram or computed tomography (CT) arthrogram could be helpful. In these cases, a visible sharp outline of the tendon sheath suggests no significant biceps inflammation; however, a negative arthrogram has been reported in up to 30% of cases with biceps pathology.^{15,16}

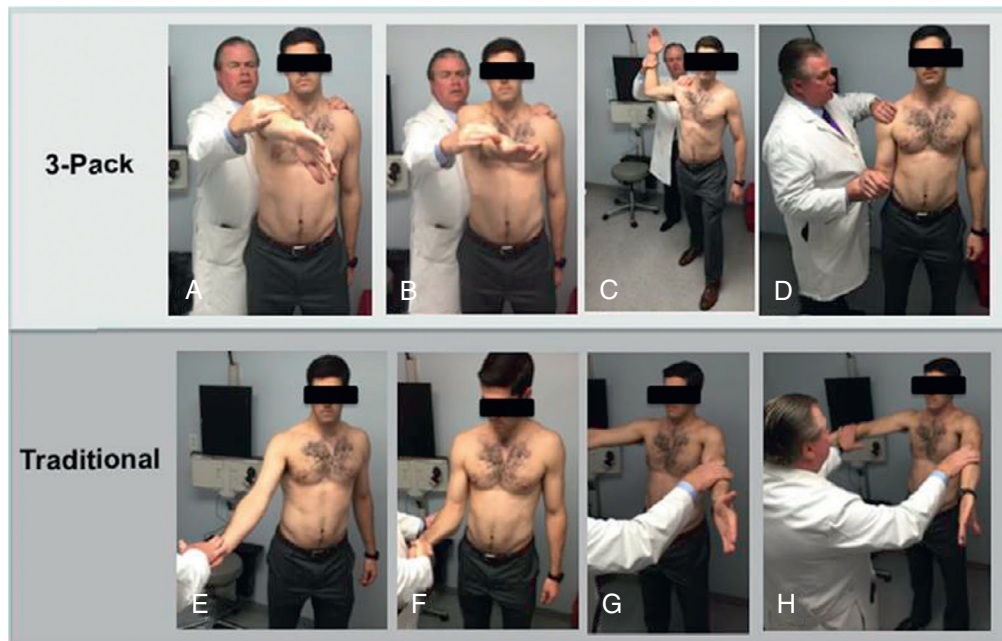


Fig. 46.1 The 3-pack examination includes (A) the active compression test with resisted elevation of the arm in pronation and (B) repeated again in supination, (C) resisted throwing maneuver, and (D) bicipital groove tenderness. Traditional examinations include (E) Speed test, (F) Yergason sign, (G) full can, and (H) empty can. (Modified from O’Brien’s article: Taylor SA, Newman AM, Dawson C, et al. The “3-Pack” examination is critical for comprehensive evaluation of the biceps-labrum complex and the bicipital tunnel: a prospective study. *Arthroscopy*. 2017;33[1]:28–38.)

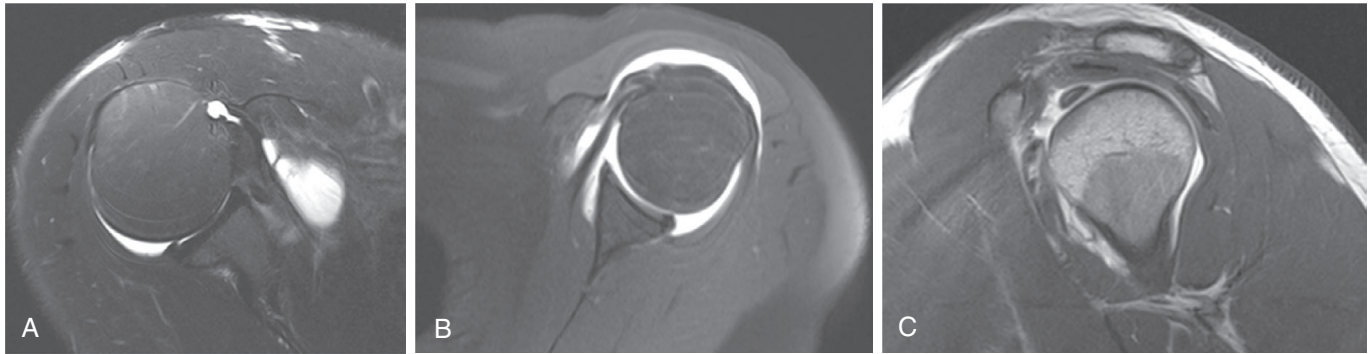


Fig. 46.2 Magnetic resonance imaging of the proximal biceps. (A) High-grade partial, near complete rupture of the long head biceps tendon at the groove entrance (axial image). (B) Biceps tendon subluxation into an upper third split in the subscapularis tendon (axial image). (C) Intrasubstance degeneration of the proximal biceps tendon within the joint (hourglass lesion) (sagittal image).

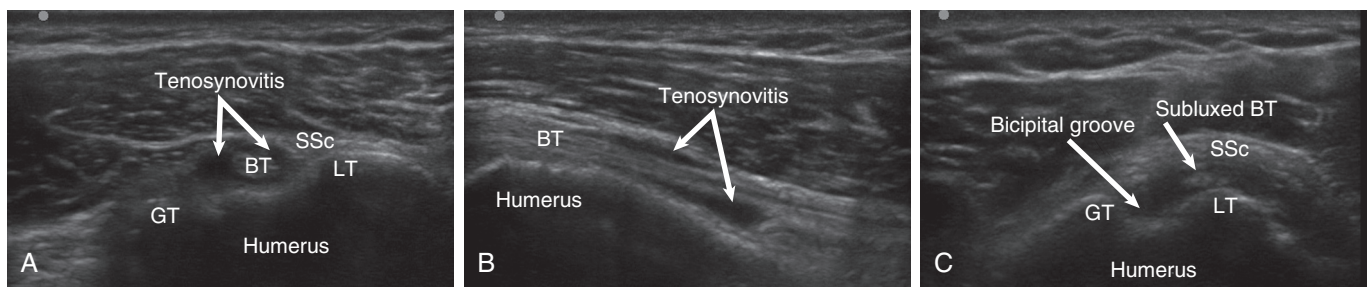


Fig. 46.3 Ultrasound images. (A) An axial view of long head biceps tenosynovitis. (B) An ultrasound longitudinal view of long head biceps tenosynovitis. (C) An ultrasound axial view of a subluxed long head biceps tendon and subscapularis tear. BT, Biceps tendon; GT, greater tuberosity; LT, lesser tuberosity; SSc, subscapularis.

MRI is the modality of choice for imaging of the biceps tendon and surrounding soft tissues structures within the shoulder (Fig. 46.2). MRI provides visualization of the biceps tendon, bicipital groove, bony osteophytes, and fluid. The biceps tendon is best visualized on the axial and sagittal oblique views. Although more sensitive than noncontrast MRIs, MRI arthrograms still show a poor concordance with arthroscopic findings concerning biceps pathology with rates of concordance reported at 34.9%.¹⁷ Complete proximal biceps ruptures are readily apparent. More subtle pathology, such as partial-thickness tears, tendinitis, and tendinopathy, are more difficult to assess. Other studies have reported sensitivity and specificity rates at 27% and 86%, respectively, regarding the ability of MRI to detect partial biceps tendon tears when compared to arthroscopy.¹⁸ MRI arthrography enhances the sensitivity and specificity in the diagnosis of biceps pathology and is the image modality of choice for evaluating SLAP tears.¹⁹ On MRI arthrography, the long head of the biceps is normally surrounded by contrast fluid and has the shape of a kidney bean on the axial view. If a subscapularis partial or full-thickness tear is suspected, the axial and sagittal oblique views should be scrutinized for potential long head of the biceps subluxation or dislocation.²⁰

Ultrasound is becoming increasingly popular in the diagnosis of musculoskeletal pathology (Fig. 46.3). Although inexpensive, it is highly operator dependent. Regarding the shoulder, ultrasound is accurate at detecting full-thickness rotator cuff tears, bicipital instability, and bicipital tendon rupture; however, it has low sensitivity (49%) when detecting partial-thickness biceps

tendon tears, and the modality is unable to assess tendon inflammation.²¹ Ultrasound has also been used for selective shoulder injections improving the accuracy of the injections specifically when injecting the bicipital tendon sheath.¹³

DECISION-MAKING PRINCIPLES

When treating any disorder, patient factors should be considered and proximal biceps tendon pathology is no different. The practitioner should consider the patient's age, activity level, body habitus/body mass index, occupation, sporting activities, comorbidities, and overall functional expectations. As discussed previously, the close proximity of potentially pathologic structures in the shoulder, such as the rotator cuff, AC joint, superior labrum, and anterior capsule, makes diagnosing and treating proximal biceps disorders challenging. A successful patient outcome necessitates the correct diagnosis of not only the proximal biceps tendon pathology but also any concomitant pathology.

Surgical intervention is reserved for cases of failed nonsurgical management consisting of rest, rehabilitation, physical therapy, nonsteroidal antiinflammatory drugs, and/or selective corticosteroid injections. Generally agreed upon indications for surgery include partial long head of the biceps tendon tears and biceps tendon subluxation with or without concomitant subscapularis tear. Relative indications include type IV SLAP tears, symptomatic type II SLAP tears in patients over the age of 35 to 40, failed SLAP repair, and chronic anterior shoulder pain from bicipital tendinitis.²²

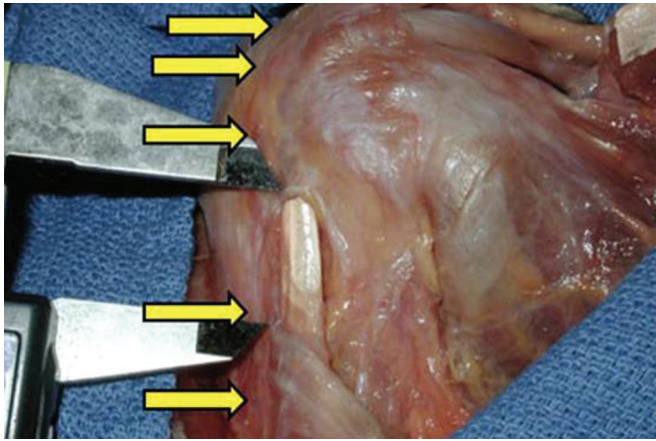


Fig. 46.4 Various tenodesis locations (arrows) of the proximal long head biceps tendon.

Biceps tenodesis is a well-described technique and can be performed arthroscopically, in conjunction with an open procedure, or stand alone as a mini open procedure. Many different techniques have been described for fixation of the tendon distally including the use of bone tunnels, keyholes, suture to decorticated bone, interference screws, and suture anchors. Interference screws have been shown to have the highest load to failure compared to suture anchors.^{23,24}

Controversy still remains over the location of the tenodesis, which includes above or in the bicipital groove, below the groove just above the superior margin of the pectoralis major tendon (suprapectoral), or in the subpectoral region just above the inferior margin of the pectoralis major tendon (Fig. 46.4). Proponents of more distal fixation (supra- or subpectoral) theorize that removing the tendon from the groove removes residual tenosynovitis potentially found within the tendon sheath.^{6,25}

TREATMENT OPTIONS

Nonoperative Management

Unless the patient has a complete tendon rupture or a high-grade partial tear at risk for complete rupture, treatment for proximal biceps tendon pathology begins with a conservative (nonoperative) management algorithm for at least 6 to 12 weeks.²⁶ A trial of rest, activity modification, oral antiinflammatory medications, and physical therapy can often lead to improvement if not resolution of symptoms.^{10,14,26} It is important that physical therapy be focused on underlying scapular disorders and concomitant shoulder pathology often seen in conjunction with biceps pathology. If pain persists, a local anesthetic combined with corticosteroid can be injected either into the glenohumeral joint or bicipital sheath, depending on the location of pathology and clinician preference (Fig. 46.5). For either location, it is imperative that the medication reach its intended location. Fluoroscopy and ultrasound are typically employed for glenohumeral joint and bicipital sheath injections, respectively, and improve both diagnostic and therapeutic accuracy, ensuring accurate placement of the injection. The success rate for biceps tendon sheath

ultrasound-guided injections has been reported at 87% compared to a 27% unguided success rate.²⁷

Surgical Management

Indications

Surgical intervention is considered in patients who have failed a conservative management algorithm. Effective surgical options are available for the management of pathologies to any portion of the LHBT. This includes SLAP lesions, partial tendon tearing greater than 25% to 50%, rupture, subluxation, or pulley lesions.²² Additional intraoperative findings that would warrant addressing the biceps tendon surgically would include a hyperemic tenosynovium, hypertrophied tendon (hourglass deformity),^{28,29} superior subscapularis tearing, or biceps chondromalacia to the humeral head.^{30,31}

Optimal surgical management of pathologies involving the LHBT remains controversial, with biceps tenotomy and tenodesis being the two most commonly performed procedures. Biceps tenotomy has the advantages of being technically less demanding to perform and does not require postoperative restrictions or immobilization. Disadvantages include cosmetic deformity (Popeye sign)³²⁻³⁴ secondary to distalization of the muscle belly, muscle cramping, and fatigue discomfort with repetitive use.³⁵⁻⁴¹ Biceps tenodesis is typically advocated by surgeons who favor the procedure in laborers, active patients, younger persons, and those concerned with cosmetic appearance.⁴² Tenodesis preserves muscle tone and prevents distal migration of the biceps, thus preserving the contour of the muscle and avoiding a cosmetic deformity.^{14,43,44} Strength and function, especially during elbow flexion and forearm supination, are maintained.⁴⁴⁻⁴⁷ While muscle cramping and spasms following tenodesis have been reported, they occur at a significantly lower rate than tenotomy.^{44,48} Disadvantages of tenodesis compared to tenotomy include a higher technical demand on the surgeon, longer postoperative restrictions and rehabilitation, and a risk of fixation failure.⁴⁹ While a traditional approach is to perform a tenotomy on patients older than 60 years of age,⁵⁰ newer studies have pointed to patient-specific preferences, such as cosmesis, need for hardware, and postoperative restrictions, as the main determinates when choosing between tenotomy and tenodesis.⁵¹

Systematic reviews have failed to demonstrate a difference in outcomes between biceps tenodesis and tenotomy.^{36,37,39,50,52} This may be due to the fact that some of these articles did not differentiate between the varying types of tenodesis, or it may be due to nonrandomization and patient selection. It also may be due to addressing associated shoulder pathologies, such as rotator cuff tears or external impingement, at the time of surgery.

Despite the lack of evidence showing superior outcomes of tenodesis over tenotomy, there has been an increase in the number of biceps tenodesis procedures being performed. In a recent study by Werner et al., there was a reported 1.7-fold increase in biceps tenodesis over tenotomy in the United States from 2008 to 2011, with an increase in the incidence of arthroscopic biceps tenodesis significantly outpacing the increase in open tenodesis.⁵³ They speculate that a portion of the overall increase in tenodesis may reflect patients with failed SLAP repairs or surgeons who

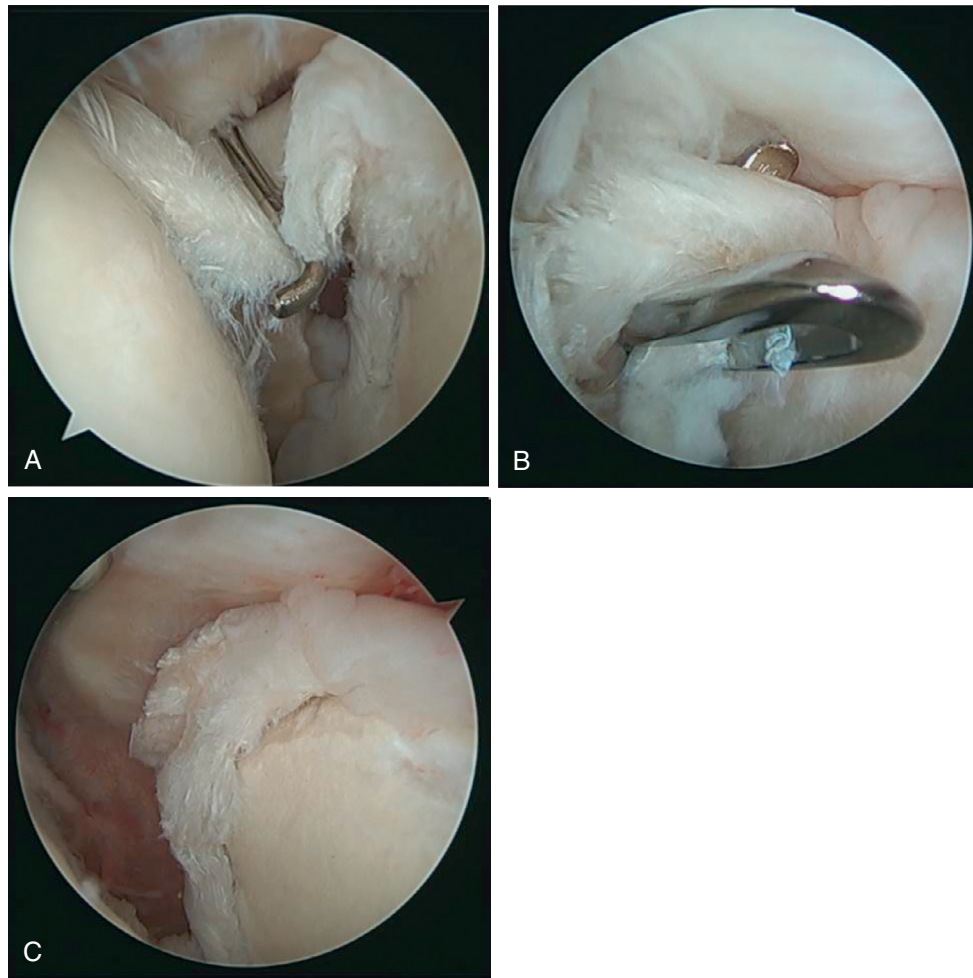


Fig. 46.5 Arthroscopic biceps tenotomy. (A) A 72-year-old male with a degenerative biceps tendon with longitudinal high-grade tearing. (B) Biceps tendon release at the superior labral attachment with a punch biter. (C) Completed biceps tenotomy.

prefer biceps tenodesis as a primary procedure for SLAP lesions based on trends in current literature.⁵⁴⁻⁵⁷ Additionally, it may be due to surgeons gaining an understanding and comfort level with performing newer techniques.

Diagnostic Arthroscopy and Biceps Tenotomy

Arthroscopic tenotomy can be performed with the patient in the lateral decubitus or beach chair position, depending on surgeon preference and concomitant procedures required. The arthroscope is introduced through a standard posterior portal. Under direct arthroscopic visualization, a spinal needle is used to localize proper placement of an anterior portal in the rotator interval. Once the anterior portal is established, a probe is inserted, and a thorough diagnostic arthroscopy of the glenohumeral joint is performed.

The LHBT is evaluated first at its attachment. The probe is used to lift the superior labrum to evaluate for a SLAP lesion. From there the biceps tendon is evaluated distally. The intra-articular tendon is inspected for tearing and tenosynovitis. The probe is used to evaluate for instability of the LHBT by attempting to displace the tendon from its groove and sling.⁵⁸ Intra-articular findings indicating possible biceps instability include

superior subscapularis tearing, anterior supraspinatus tearing, or antero-superior humeral head chondromalacia.^{30,31} The probe is further utilized to pull the biceps tendon into the joint. Forward shoulder elevation and elbow flexion can provide further excursion of the biceps tendon.^{59,60} The distal tendon is inspected for inflammation, tearing, or hypertrophy (hourglass deformity).²⁹ The biceps pulley, a capsuloligamentous complex formed by a coalescence of fibers form the coracohumeral ligament, superior glenohumeral ligament, and portions of the subscapularis and supraspinatus tendons, functioning to stabilize the LHBT within the proximal portion of the groove,⁴ is inspected for pathology. To aid in arthroscopic evaluation, some advocate the use of a 30-degree scope via the lateral portal or a 70-degree scope via the posterior portal.⁶¹

After a diagnostic arthroscopy the biceps tenotomy can be performed. Viewing from the posterior portal, the biceps is tenotomized using a punch biter or electrocautery device via the anterior portal to separate the tendon from its attachment to the superior labrum and supraglenoid tubercle (see Fig. 46.5). Careful attention is made not to violate the labrum. Any residual stump is gently débrided using a shaver. The elbow can be extended and the biceps tendon observed as it retracts. If the

biceps tendon does not retract into the bicipital groove, it may be due to hypertrophied proximal tendon. This will need to be excised. If it still does not retract, the bicipital sheath may need to be released arthroscopically, depending on biceps pathology observed during diagnostic arthroscopy and the location of the patient's pain prior to surgery.

Biceps Tenodesis

Tenodesis of the LHBT can be performed in a variety of ways that include arthroscopic^{28,33,45,62,63} or open^{10,44,55,64} technique and can be positioned high at the entrance of the bicipital groove at the chondral junction,^{49,61,65} in the suprapectoral location just proximal to the pectoralis major tendon,^{42,44,45} in a subpectoral location at, or distal to, the pectoralis major tendon,^{10,44,55,64} or in other positions, including the conjoint tendon or soft tissue tenodesis sites.^{26,66,67} Regardless of location, there is an increasing recognition that restoration of the anatomic length-tension relationship of the LHBT is a critical aspect of the tenodesis procedure.^{42,63,68,69}

Arthroscopic Biceps Tenodesis

Biceps tenodesis can be performed entirely arthroscopically without the need for a separate larger incision. This minimizes the risk of infection, which is a concern with open subpectoral tenodesis that uses an incision near the axillary fold. A disadvantage of arthroscopic compared to open subpectoral tenodesis is an increased risk of postoperative stiffness.⁷⁰ Arthroscopic biceps tenodesis can be performed in the beach-chair or lateral decubitus positions. After a diagnostic arthroscopy and prior to release from its superior labral attachment, the tendon is typically tagged percutaneously to prevent retraction and preserve the length-tension relationship.^{42,44,68,69,71}

Two main locations for arthroscopic tenodesis include a high intertubercular groove entrance site at the chondral junction, and a low suprapectoral site just proximal to the pectoralis major tendon. Advocates for tenodesis at the chondral junction note that the interference screw can be incorporated as a suture anchor during rotator cuff repair, a procedure commonly performed in conjunction with biceps tenodesis.⁶¹ It also leaves a substantial amount of tendon proximal to the pectoralis major, which allows for revision tenodesis should it be necessary. In some studies, this location results in a low surgical revision rate of 0.4%, a low rate of residual pain, and a significant improvement in outcome scores.⁶⁵ However, there are those who feel that a tenodesis performed at the chondral junction does not address bicipital tunnel disease, which can be a source of residual groove pain, and instead advocate for a suprapectoral tenodesis site.^{33,63} This location places the tenodesis tunnel distal to the bicipital groove, which allays concerns about the groove as a source of pain after tenodesis.⁶²

Arthroscopic tenodesis fixation, regardless of location, can be performed with a variety of methods, most commonly with an interference screw^{33,45,61,72} or suture anchor.^{47,73,74} In a biomechanical study by Richards and Burkhart,⁷⁵ a biceps tenodesis with interference screw provided greater fixation strength than the same procedure performed with a double suture anchor technique.

Open Biceps Tenodesis

Open biceps tenodesis is typically performed at the subpectoral location. The procedure is less technically demanding than arthroscopic tenodesis and removes the majority of the LHBT and associated tenosynovium from within the groove, thus eliminating a potential source for postoperative pain.^{44,60,64} In a biomechanical study using matched cadavers, open subpectoral tenodesis had a significantly increased ultimate load to failure when compared with an arthroscopic suprapectoral technique using an interference screw.⁷⁶ Disadvantages of the open subpectoral technique include the need to make an incision near the axilla, which has the potential for wound issues, as well as the removal of the majority of the LHBT, which leaves little usable tendon should a revision be necessary.

Open biceps tenodesis can be performed in the lateral decubitus or beach-chair position depending on surgeon preference. Diagnostic arthroscopy is performed first. Prior to tenotomy, a tag suture is typically placed in the biceps tendon to prevent overretraction and to aid in tendon identification. After tenotomy, the open portion of the case is begun. A 3 cm longitudinal incision is made at the inferior border of the pectoralis major tendon near the axillary fold on the anteromedial aspect of the upper arm. The tendon is retrieved through the wound and whipstitched. Either an interference screw,¹⁰ cortical button, or bone tunnel³² is typically used for fixation. Compared with suprapectoral biceps tenodesis with interference screw, subpectoral tenodesis with interference screw may have an increased risk of proximal humerus fracture through the humeral socket for the screw.^{77,78}

Other Techniques

Soft tissue tenodesis, as described by Sekiya et al., involves securing the biceps tendon to the soft tissues in the rotator interval.⁵⁹ In this technique, a spinal needle is percutaneously placed through the lateral aspect of the rotator interval and through the biceps tendon approximately 1 to 2 cm distal to its origin on the superior labrum. Suture is shuttled through and retrieved through the anterior portal. The process is repeated, passing suture distal to the first but proximal to the bicipital groove. The scope is introduced into the subacromial space and the suture ends retrieved through a cannula and tied. Tendon proximal to the tenodesis site is then excised.

Originally described by Froimson, the keyhole technique involves first performing a proximal tenotomy.⁷⁹ The tendon is then retrieved through an open suprapectoral approach and the proximal end balled-up. In the bicipital groove a bone tunnel with distal trough (keyhole) is created. The balled-up portion of the tendon is inserted into the bone tunnel. The distal, tubular portion of the tendon is slide distally in the trough and is secured in place by the balled-up tendon that is deep in the canal and too wide for the trough.^{80,81}

Another less commonly used procedure includes the arthroscopic transfer of the LHBT to the conjoint tendon.^{26,67} With this technique, the LHBT is released from its origin, transferred to the subdeltoid space and sutures to the conjoint tendon in a side-to-side manner with nonabsorbable sutures.

Authors' Preferred Techniques

Arthroscopic Suprapectoral Biceps Tenodesis

The biceps tendon is tagged while viewing arthroscopically by percutaneously inserting a spinal needle just lateral to the anterior portal and piercing the biceps tendon (Fig. 46.6). The arthroscope is placed in the subacromial space via the posterior portal. Using a spinal needle and outside-in technique, a standard lateral portal is established. A combination of a shaver and radiofrequency ablator are used to remove the subacromial bursa in order to aid in visualization.

The arthroscope is then moved to the lateral portal. With the radiofrequency ablator in the anterior portal, the arm is forward flexed to approximately 80 degrees and slightly externally rotated while dissection is continued anterior and distal to identify the biceps tendon in between the distal margin of the subscapularis tendon and proximal margin of the pectoralis major tendon.²⁵ Starting lateral and working medial, the biceps tendon is liberated from its sheath. Once mobilized, a spinal needle is used to localize for a distal anterolateral portal directly over the biceps tendon and just proximal to the pectoralis major tendon. We have found that identifying a point distal and equidistant to the anterior and lateral portals creates an equilateral triangle and is a typical location for creating an anterolateral portal.

After needle localization and establishment of an anterolateral portal, a cannula is introduced. A suture-passing device is introduced through the cannula and used to pass a heavy, nonabsorbable suture in luggage-tag fashion. Each limb is additionally passed through the tendon, one immediately proximal and the other distal to the first suture pass. At the site where the tendon is sutured, a guide pin is placed perpendicular to the humerus and in the bicipital groove. The near cortex is then reamed with a cannulated reamer to the appropriate diameter given the size of the tendon and desired interference screw. We most commonly ream to the same size of the interference screw (line to line), which is typically 4.5–7 mm in diameter. It is important to ensure that the reamed depth is enough to accommodate the screw. A radiofrequency ablator is used to tenotomize the tendon just proximal to the suture. The suture limbs are passed through the fork-tipped interference screw and the tendon is guided into the bone socket and screwed into place. Care is taken not to countersink the screw as this can result in reduced fixation. The suture is subsequently sewn over the top to reinforce the fixation. The arthroscope is then reinserted into the glenohumeral joint and the residual biceps tendon is tenotomized at its origin and removed with a traumatic grasper.

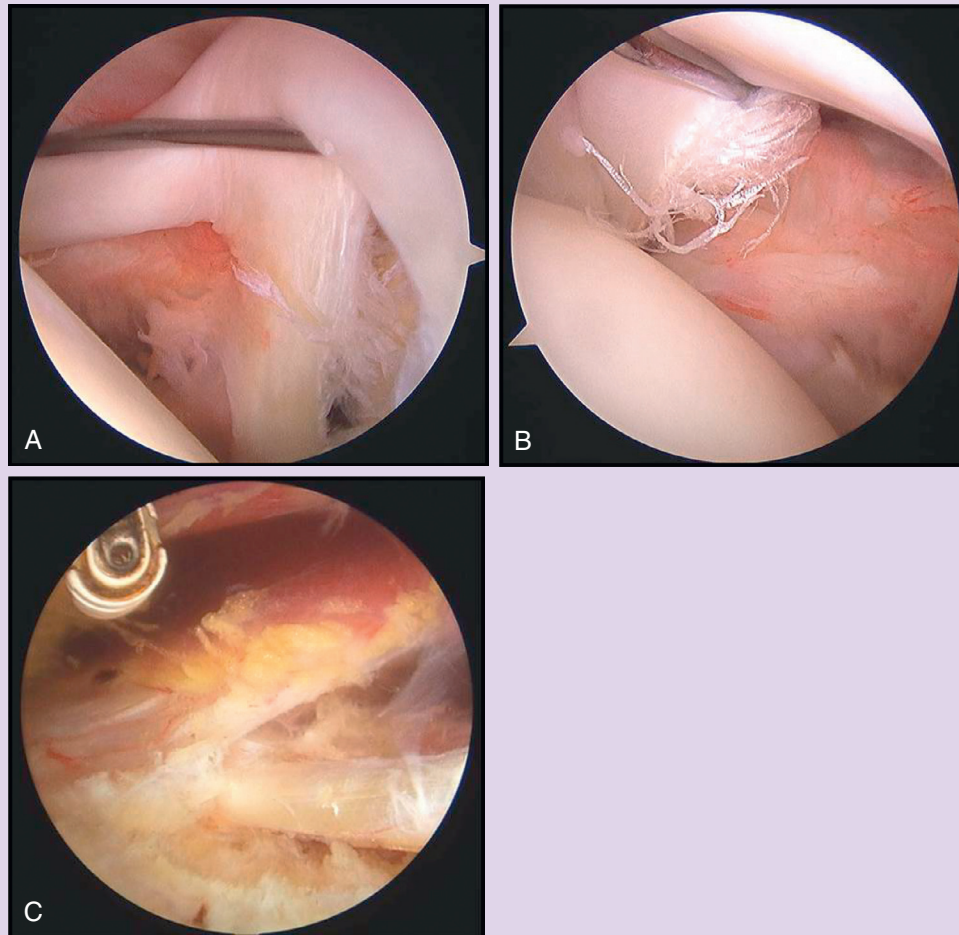


Fig. 46.6 Arthroscopic suprapectoral biceps tenodesis. (A and B) A 56-year-old male with a type 2 superior labral from anterior to posterior tear and a high-grade partial long head of the biceps tendon (LHBt) tear at the groove entrance. (C) Identification of the LHBt at the upper border of the pectoralis major tendon (viewing from direct lateral portal, working from low anterior “tenodesis” portal).

Continued



Authors' Preferred Technique—cont'd

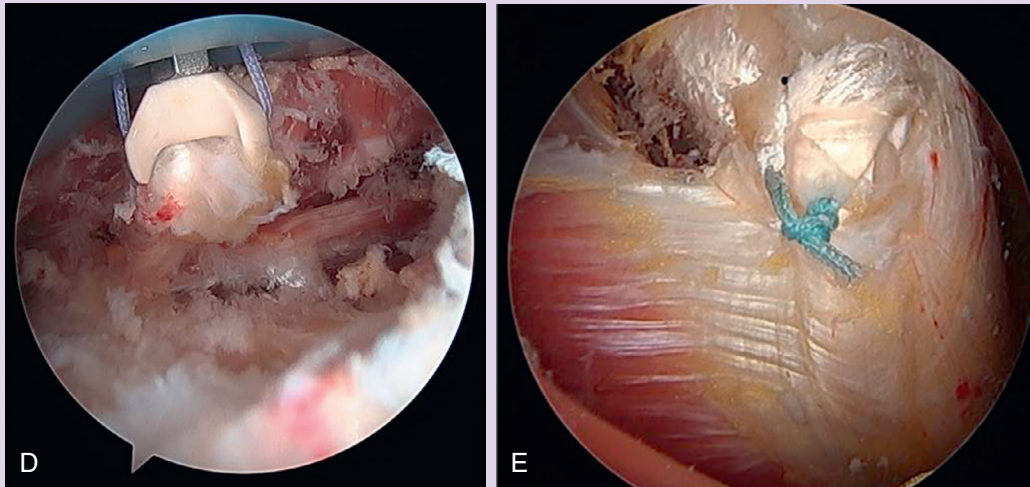


Fig. 46.6, cont'd (D) LHBt has been secured with a suture and then tenotomized at the planned location for tensioning; the humeral tunnel has been prepared; and the tendon is attached to an interference screw for tenodesis. (E) Final tenodesis at the upper border of the pectoralis major tendon, secured with an interference screw and suture back-up.

Open Subpectoral Biceps Tenodesis

After a diagnostic shoulder arthroscopy, a tag suture is placed into the biceps tendon under arthroscopic visualization and a tenotomy is performed (Fig. 46.7). Subpectoral tenodesis is performed in an open fashion through a longitudinal incision at the lower border of the pectoralis major muscle as described by Mazzocca et al.¹⁰ An ~3 cm vertical incision is made just lateral to the axillary fold, with 1 cm over the pectoralis major tendon and 2 cm distal to the inferior border. The skin is incised sharply and blunt dissection is used to identify the fascia overlying the biceps brachii. The pectoralis major tendon can be palpated to ensure the approach is along its inferior border. The fascia is carefully incised longitudinally. Blunt dissection is used to locate the LHBt. The tag suture can be utilized to pull tension on the biceps tendon to facilitate identification. The surgeon's finger or a right-angled clamp is used to deliver the tendon from the wound.

Starting at the musculotendinous junction and working 2 cm proximally, the biceps tendon is whipstitched using heavy, non-absorbable suture. Excess tendon is transected using a knife. The suture limbs are passed through both holes of a two-hole cortical button but in opposite directions. The suture limbs are pulled to ensure the button slides freely. A pointed Hohmann retractor is placed around the lateral border of the humerus to retract the pectoralis major superiorly and laterally. A Chandler retractor is placed around the medial aspect of the humerus to retract the coracobrachialis and short head of the biceps medially while at the same time protecting the neurovascular structures. In a cadaveric study, it was reported that the average distance of the musculocutaneous nerve to the musculotendinous junction of the long head of the biceps is 2.6 cm. Controlled retraction with the Chandler retractor should be employed to prevent traction injuries to surrounding nerves. The bicipital groove should be

able to be visualized and palpated. If further retraction is needed, a Sauerbruch or Army-Navy retractor can be used to pull the inferior border of the pectoralis major proximally.

The osseous bed is prepared with a periosteal elevator. A spade-tip drill bit specific to the cortical button implant is drilled unicortically, centered in the groove and approximately 3 cm proximal to the inferior border of the pectoralis major insertion. The button is deployed within the humeral canal and flipped by toggling the suture limbs. The length-tension relationship of the biceps is checked. A free needle is used to pass one suture limb through the proximal tendon and the suture limbs are tied.

POSTOPERATIVE MANAGEMENT

Biceps Tenotomy Rehabilitation Protocol

The goals of rehabilitation following biceps tenotomy are to control pain and inflammation in the acute phase while attempting to restore range of motion (ROM) and function. Patients are placed into a simple sling for comfort following surgery and are instructed to discontinue it within the first 1 to 2 weeks. Rotator cuff isometrics and scapular exercises are initiated immediately after surgery without any limitation to ROM. After 2 weeks patients can begin isotonic rotator cuff and scapulothoracic strengthening and advance as tolerated. Starting 6 weeks after surgery, patients can begin conventional weight lifting with machine weights and progress slowly to free weights. If the patient is free of pain and has reached their baseline strength and motion, they can return to sport or activity.

Biceps Tenodesis Rehabilitation Protocol

Whether the biceps tenodesis was performed arthroscopically or open, the rehabilitation protocol is the same. It is imperative that the biceps tendon be protected to allow bone-tendon healing.^{64,82} Biceps loading is not allowed for the first 12 weeks

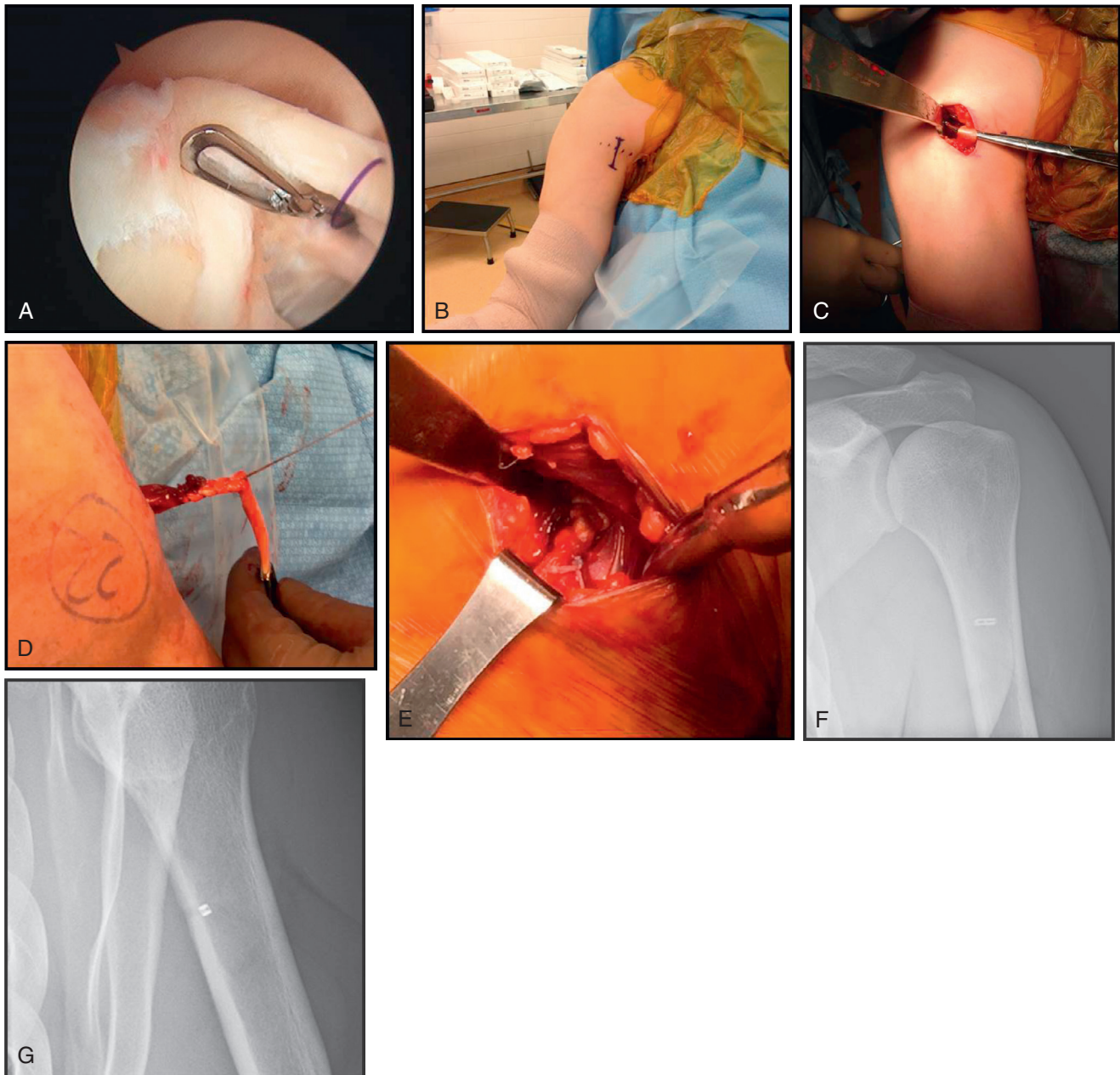


Fig. 46.7 Open subpectoral biceps tenodesis. (A) In a 43-year-old male with a type 2 superior labral from anterior to posterior lesion, the long head of the biceps tendon (LHBT) has been tagged with polydioxanone suture and is tenotomized. (B) Typical incision for an open subpectoral biceps tenodesis positioned lateral to the axilla with one-third above the lower edge of the pectoralis major tendon, two-thirds below. (C) LHBT is secured in the subpectoral region and externalized. (D) A tendon-grasping stitch has been placed securing the tendon along the 20 mm segment above the muscle/tendon junction. (E) The tenodesis is completed with the tendon secured to the anterior humerus using a unicortical button fixation. (F and G) Final radiographs of an open subpectoral biceps tenodesis.

following surgery. Throwing or overhead sports are not allowed until week 20.

Phase I: Protective Phase (Day 1 to Week 6)

The goals of phase I are to control pain, swelling, and inflammation, and to restore ROM. The arm is placed into an ultrasling for the first 6 weeks. Immediately after surgery, passive and gentle active assisted ROM exercises are initiated under the guidance of a physical therapist and are limited to flexion and scaption

to 90 degrees, ER to 30 degrees, and internal rotation (IR) to 45 degrees for the first 4 weeks. Starting week 5, ROM is increased with flexion and scaption to full, ER to 50 degrees, and IR to 60 degrees.

Phase II: Moderate Protection Phase (Weeks 7 to 12)

The goals of phase II are to initiate active ROM and increase muscle endurance. The sling is discontinued. Both active and

TABLE 46.2 Clinical Outcomes of Proximal Long Head Biceps Tenodesis

Procedure Type	Authors	Study Design	Patients	LHBT Condition	Concomitant Procedures
Arthroscopic/anchor	Checchia et al. (2005) ⁴⁶	Level IV retrospective case series	<i>n</i> = 15; mean age, 62 years; range, 41–80 years	Subluxated, 13% (2/15); dislocated, 40% (6/15); prereductured, 47% (7/15)	RCT, 100% (15/15); acromioplasty, 67% (10/15); SLAP, 7% (1/15)
Arthroscopic/suture anchor through subclavian portal	Nord et al. (2005) ⁴⁷	Level IV prospective case series	<i>n</i> = 10; mean age 60 years; range 41–77 years	NR	RCT (all intra-articular) and/or labral tears 100% (10/10); acromioplasty 40% (4/10); impingement syndrome 100% (10/10)
Arthroscopic/suture anchor	Franceschi et al. (2007) ⁵²	Level IV retrospective case series	<i>n</i> = 22; mean age, 60 years; range, 40–81 years	Subluxated, 32% (7/22); dislocated, 32% (7/22); prereductured, 36% (8/22)	RCT 100% (22/22); other concomitant repairs NR
Arthroscopic/interference screw	Boileau et al. (2002) ⁴⁵	Level III prospective, nonrandomized case series	<i>n</i> = 43; mean age, 63 years; range, 25–78 years	Subluxated, 26% (11/43); dislocated, 30% (13/43); prereductured, 35% (15/43); tenosynovitis, 9% (4/43)	RCT, 86% (37/43); SLAP, 14% (6/43)
Arthroscopic/interference screw	Koh et al. (2010) ⁹⁰	Level II cohort study	<i>n</i> = 43; mean age, 61 years; range, 55–77 years	Subluxated/dislocated, 9% (4/43); prereductured, 12% (5/43); severe tenosynovitis, 7% (3/43)	RCT, 79% (34/43); SLAP, 44% (19/43); SAD, 5% (2/43); DCE, 12% (5/43)
Open/suture anchor	Warner et al. (2001) ⁹¹	Level IV retrospective case series	<i>n</i> = 19; mean age, 58 years; range, 36–72 years	Subluxated, 100% (19/19); prereductured, 74% (14/19)	RCT, 100% (19/19); acromioplasty, 79% (15/19)
Mini open/interference screw	Mazzocca et al. (2008) ⁶⁴	Level IV retrospective case series (single surgeon)	<i>n</i> = 41; mean age, 50 years; range, NR	NR	RCT, 59% (24/41); Bankart w/ anterior capsulorrhaphy, 5% (1/41); SAD, 20% (8/41); DCE, 10% (4/41); acromioplasty, 5% (2/41); SLAP, 5% (2/41); lipoma excision, 5% (1/41)
Open/interference screw and suture anchor	Millett et al. (2008) ⁸⁶	Level IV retrospective case series (2 surgeons)	<i>n</i> = 88; interference screw 34, suture anchor 54; mean age, 51 years; range, 22–77 years	Subluxated (NR)	RCT, 73% (64/88); DCE, 14% (12/88); acromioplasty, 47% (41/88); capsular reconstruction, 9% (8/88)
Open/interference screw	Nho et al. (2010) ⁸⁵	Level IV retrospective case series	<i>n</i> = 13; mean age, 52 years; range, 32–65 years	Subluxated, 100% (13/13)	RCT, 100% (13/13); acromioplasty, 100% (13/13); DCE, 31% (4/13)

ASES, American Shoulder and Elbow Surgeons; CM, Constant Murley Functional Assessment of the Shoulder; DCE, distal clavicle excision; LHBT, long head of the biceps tendon; MC, modified CM; MRI, magnetic resonance imaging; NR, not reported; RC, rotator cuff; RCT, randomized controlled trial; ROM, range of motion; SAD, subacromial decompression; SANE, single assessment numeric evaluation; SLAP, superior labral anterior to posterior; SST, simple shoulder test; UCLA, University of California–Los Angeles; VAS, visual analogue scale.

Follow-Up	Outcome Measures	Positive (Excellent/Good Outcome)	Negative (Fair/Poor Outcome)	Overall Results
Mean, 32 months; range, 20–67 months	UCLA; ROM	93% (14/15); improved scores; improved ROM	7% (1/15); weakness and residual pain; MRI revealed severe atrophic RC muscles, but RC suture and tenodesis were intact	Concomitant repair revealed satisfactory results with no significant complications
Mean, 24 months; range, NR	UCLA	90% (9/10); improved scores	10% (1/10); 1 patient had an adhesive capsulitis postoperative complication at 2-year follow-up	Suture anchor placement slightly proximal to bicipital groove was effective for LHBT tenodesis with concomitant intra-articular RCTs
Mean, 47 months; range, 36–59 months	UCLA; ROM	100% (22/22); improved scores; improved ROM	0%	No patients rated surgery as unsuccessful; no patients had bicipital pain, Popeye sign, or daily activity limitations
Mean, 17 months; range, 12–34 months	CM; ROM strength	95% (41/43); improved scores; no loss of elbow movement, and biceps strength was 90% of the other side	5% (2/43); rupture of tenodesis; MRI revealed tight fixation of the biceps tendon in the humeral socket	Good clinical results with isolated pathologic biceps tendon or cuff repair; very thin, fragile, almost ruptured biceps tendon is the technical limit to the arthroscopic approach
Mean, 28 months; range, 24–35 months	Patient satisfaction; ASES; CM	84% (36/43); improved scores	16% (7/43); 2 patients complained of arm pain at resisted flexion; 4 patients had Popeye deformity	Suture anchor tenodesis of LHBT is associated with <10% incidence of Popeye deformity
Mean, 40 months; range, 24–75 months	Patient satisfaction; MC	42% (8/19); improved scores	58% (11/19)	Although prior surgery did not directly correlate with Constant score, delayed repair of more than 6 months of symptoms resulted in negative outcomes; 9/19 patients had prior failed RC surgeries, which may have resulted in less favorable outcomes
Complete follow-up at minimum 1 year after surgery (mean, 29 months; range, 12–49 months)	Rowe; ASES; SST; CM; SANE	100% (41/41); improved scores	0%	All clinical outcome measures (Rowe, ASES, SST, CM, SANE scores) demonstrated a statistically significant improvement at follow-up when compared with preoperative scores for all 41 patients; 1 failure was due to pull-out of tendon from bone tunnel resulting in Popeye deformity; patients with coexistent RC lesions had less favorable outcomes (lower ASES, SST scores)
Mean, 13 months; range, 3–25 months	Patient satisfaction; SES; VAS; MC	87% (48/55) (subjective outcome only collected for n = 55); mean scores improved (scores collected for all n = 88)	5% (4/88); 1 patient indicated shoulder was worse after surgery and 1 patient did not return to normal activities	No failures of fixation; no complications after surgery; 5 patients reported persistent groove tenderness; 2 patients reported persistent spasm
Mean, 35 months; range, 14–52 months)	Patient satisfaction; ASES; SST; VAS	100% (13/13); improved scores	0%	Arthroscopic repair of anterosuperior RC tears with open biceps tenodesis significantly improved clinical outcomes scores, pain relief, and shoulder function

passive ROM are increased with a goal of full motion by week 10. Starting week 10, submaximal isometrics and active ROM are initiated. The patient may also begin more aggressive exercises for rotator cuff and scapulothoracic musculature.

Phase III: Minimum Protection Phase (Weeks 13 to 20)

The goals of phase III are for full, nonpainful active and passive ROM, restoration of muscle strength, power and endurance, and gradual initiation of functional activities. Isotonic elbow flexion and forearm supination are initiated along with light plyometric activities.

Phase IV: Advanced Strengthening Phase (Weeks 21 to 26)

The goals of phase IV are to increase functional activities and return to sport. Overhead athletes may slowly begin their sport specific activities.

RESULTS

Regardless of technique, most articles demonstrate good to excellent outcomes following biceps tenodesis with relief of pain and return of function (Table 46.2).^{22,53,60,64,83,84}

There is sufficient evidence in the literature to support performing open subpectoral biceps tenodesis for the treatment of pathology to the LHBT. Mazzocca et al. evaluated outcomes after open subpectoral biceps tenodesis with an interference screw. They reported improved patient reported outcomes in 41 patients at an average follow-up of 29 months.⁶⁴ In a similar study, Nho et al. obtained outcomes of open subpectoral tenodesis with interference screw in concomitant arthroscopic rotator cuff repairs and found that all 13 patients had significant clinical improvement in outcome scores at 1 year after surgery.⁸⁵ Millet et al. also reported successful results in a study comparing open subpectoral tenodesis fixation with interference screw to suture anchor. Both groups demonstrated significant improvement in outcomes scores at 13 months postoperatively, but a higher rate of bicipital groove pain in the suture anchor subset compared with the interference screw subset.⁸⁶

Clinical outcomes of arthroscopic biceps tenodesis are not as well reported as those of open subpectoral tenodesis. However, a reasonable amount of current literature demonstrates improved outcomes and patient satisfaction. Boileau et al. reported on 43 patients who underwent arthroscopic biceps tenodesis with interference screw. They reported a significant improvement in postoperative Constant scores.⁴⁵

Current literature also demonstrates comparable results after arthroscopic tenodesis to open subpectoral biceps tenodesis. In a study comparing arthroscopic suprapectoral and open subpectoral biceps tenodesis, Werner et al. showed that arthroscopic tenodesis had a higher rate of stiffness.⁷⁰ However, in a follow-up study comparing 2-year results, that stiffness resolved as both arthroscopic suprapectoral and open subpectoral tenodesis demonstrated no significant differences in clinical and functional outcomes, including ROM, between techniques.⁸⁷

COMPLICATIONS

There is a low incidence of complications following biceps tenodesis. These complications include hematoma, seroma, wound infection, humeral fracture, persistent bicipital pain, shoulder stiffness, fixation failure, cosmetic deformity, and neurovascular injury, with the latter being less common in arthroscopy due to the close proximity of the open approach to neurovascular structures.^{84,88,89} In a review of 353 patients who underwent open subpectoral biceps tenodesis with interference screw fixation, there were seven reported complications (2.0%). These included persistent bicipital pain, failed fixation, wound infection, musculocutaneous neuropathy, proximal humerus fracture, and complex regional pain syndrome.⁸⁴

Biceps tenotomy is not without complications. These include persistent bicipital pain, cosmetic deformity, muscle cramping, and fatigue discomfort with repetitive use.³⁵⁻⁴¹

FUTURE CONSIDERATIONS

Recognition of optimal location for tenodesis, whether at the articular margin, suprapectoral, or subpectoral regions, is an important area for future clinical investigation. Biceps tenodesis is commonly performed in conjunction with rotator cuff repair, which has a confounding effect when comparing outcomes after surgery. To date, there are no randomized trials collecting prospective data comparing isolated tenodesis at these locations. As we still try to gain an overall understanding of the function of the LHBT as it relates to the mechanics of the glenohumeral joint, we will perhaps develop new techniques for tendon restoration in the setting of tendon pathology.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Werner BC, Burrus MT, Miller MD, et al. Tenodesis of the long head of the biceps: a review of indications, techniques, and outcomes. *JBJS Rev.* 2014;2(12).

Level of Evidence:

Basic science review, operative technique

Summary:

This article gives a comprehensive review of arthroscopic and open, suprapectoral and subpectoral biceps tenodesis. It discusses outcomes and provides a review of the current literature.

Citation:

Mazzocca AD, Cote MP, Arciero CL, et al. Clinical outcomes after subpectoral biceps tenodesis with an interference screw. *Am J Sports Med.* 2008;36(10):1922–1929.

Level of Evidence:

IV, retrospective case series

Summary:

Clinical outcomes were reported to be good to excellent in all 41 patients undergoing subpectoral biceps tenodesis. Patients with coexisting rotator cuff lesions had less favorable outcomes.

Citation:

Nho SJ, Reiff SN, Verma NN, et al. Complications associated with subpectoral biceps tenodesis: low rates of incidence following surgery. *J Shoulder Elbow Surg.* 2010;19(5):764–768.

Level of Evidence:

IV, case series, treatment study

Summary:

Outcomes in 373 patients undergoing open subpectoral biceps tenodesis by two fellowship-trained orthopaedic surgeons at a single institution.

Citation:

Nho SJ, Strauss EJ, Lenart BA, et al. Long head of the biceps tendinopathy: diagnosis and management. *J Am Acad Orthop Surg.* 2010;18:645–656.

Level of Evidence:

Basic science review, operative technique

Summary:

This article provides a comprehensive review of biceps tendinopathy, starting with anatomy, pathophysiology, and clinical diagnosis before discussing techniques and results for tenotomy, open and arthroscopic tenodesis.

Citation:

Brady PC, Narbona P, Adams CR, et al. Arthroscopic proximal biceps tenodesis at the articular margin: evaluation of outcomes, complications, and revision rate. *Arthroscopy.* 2015;31:470–476.

Level of Evidence:

IV, therapeutic case series

Summary:

This article retrospectively reviews 1083 patients who underwent an arthroscopic biceps tenodesis at the articular margin by interference screw fixation.

Citation:

Elser F, Braun S, Dewing CB, et al. Anatomy, function, injuries, and treatment of the long head of the biceps brachii tendon. *Arthroscopy.* 2011;27(4):581–592.

Level of Evidence:

V

Summary:

Current review of the anatomy and biomechanical properties of the long head of the biceps tendon and evidence-based approach to current treatment strategies for disorders to the biceps tendon.

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